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# Synthesis of 3D MRI Brain Images With Shape and Texture Generative Adversarial Deep Neural Networks

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**ABSTRACT** Generative Adversarial Networks (GAN) are emerging as an exciting training paradigm which promises a step improvement to the impressive feature learning capabilities of deep neural networks. Unlike supervised learning approaches, GAN learns generalizable features without requiring labeled images to achieve new capabilities like distinguishing previously unseen anomalies, creating novel instances of data and factorizing learned features into explainable dimensions in fully unsupervised fashion. The advanced feature learning property of GAN will enable the next generation of computational image understanding tasks. However, GAN models are difficult to train to converge towards good models, especially for high resolution and high dimensional datasets like image volumes. We develop a GAN approach to learn a generative model of T1-contrast 3D MRI image volumes of the healthy human brain by training on 1112 MRI images from the Human Connectome Project. Our method utilizes a first unconditional Super-Resolution GAN, dubbed the shape network, to learn the 3D shape variations in adult brains and a second conditional pix2pix GAN, dubbed the texture network, to upgrade image slices with realistic local contrast patterns. Novel 3D MRI images are synthesized by first applying the 3D voxel-wise deformation map which is generated from the shape network to deform the Montreal Neurological Institute (MNI) brain template and subsequently performing style transfer on axial-wise slices using the texture network. The Maximum Mean Discrepancy (MMD) and Multi-scale Structural Similarity Index Measure (MS-SSIM) scores of MRI image volumes synthesized using our GAN approach are competitive with state-of-art GAN methods. Our work establishes the feasibility of an alternative approach to high-dimensional GAN learning - splitting the type of information content learned among several GANs can be an effective form of regularization and complementary to latent code shaping or super-resolution approaches in state-of-the-art methods.

**INDEX TERMS** Generative adversarial networks, MRI brain images, brain morphometry, deep neural networks, computational brain anatomy.

## I. INTRODUCTION

Generative Adversarial Networks (GAN) have emerged as an alternative and powerful paradigm for training deep neural networks (DNN) in unsupervised fashion [1], [2] through convincing preliminary demonstrations in various novel applications. GAN is not the only credible approach to deep unsupervised learning; the field will continue to benefit from

cross-fertilization of ideas between other competitive and emerging methodologies like Variational Autoencoders [3] and contrastive learning [4]. Yet the GAN concept has significantly contributed towards expanding the usefulness of DNNs beyond image and pixel-level classification/regression tasks because GANs learn complex image features with improved generalization properties. GANs can make decisions on previously unknown or unseen data classes [5] and synthesize novel exemplars with features similar to the training dataset [6]. The generative modeling ability of GANs

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has been effectively applied to anomaly detection tasks and to produce creative and artistic content. Innovations like StyleGAN [7], [8] demonstrate that GAN can learn feature representations in an intuitively factorizable manner. From these early achievements, GAN is evolving to be an enabling technology for next-generation computational image understanding because GAN provides a method for effective, explainable learning of complex and salient image features in a totally unsupervised manner. We argue that developing GAN models of important datasets will be a critical pre-requisite and building block for future applications relying on automated computational image understanding.

Images of the brain are an important dataset and understanding the brain has emerged as the next big science with global interest [9]. Brain images may be acquired from various instrument modalities such as magnetic resonance imaging (MRI), positron emission tomography (PET) and x-ray computed tomography (CT), which measure either anatomical features or functional activity. Functional and anatomical brain image stacks are high dimensional with multi-scale spatial and temporal organization [10] that is only beginning to be understood. Unlike man-made objects, brain anatomy comprises regions of differentiated neurons with smooth and undulating boundaries that are not easily nor efficiently represented by established feature descriptor methodologies like SIFT [11]. Both anatomical and functional connectivity of neurons also exhibit a multi-layered, hierarchical and stereotypical small-world network organization with subtle yet significant inter-individual variations [12]. Image stacks from different modalities also contain potential correlations with an untapped wealth of clinical information [13]. To address these challenges, the neuroscientific community has developed novel and unique algorithms to extract and compare salient shape and activity features from both functional and anatomical brain image datasets [14], [15]. These approaches have underscored the remarkable progress in understanding human brain organization and its relation to disease in the past 2 decades and are based on the key idea of referencing a template brain [16]. A variety of established template brains [17], [18] have been developed for anatomical and functional datasets i.e. the Montreal Neurological Institute ICBM-152 [19]. Brain image stacks from subjects must be individually deformed [20], [21] to achieve a dense, nonlinear spatial correspondence with the template, after which, statistical comparisons may be made. Besides subject to template registrations, it is also common to perform image registrations between image stacks acquired using different instrument modalities such as between T1-contrast anatomical MRI with T2-contrast functional MRI or diffusion-weighted MRI. Brain registration algorithms in established brain processing pipelines [15], [22] apply a two-step transformation process namely a global affine alignment and a local nonlinear registration. State-of-art nonlinear registration algorithms apply a topology-preserving diffeomorphic transform, which iteratively optimizes an energy function and are computationally

demanding. Examples of popular implementations for brain registration include Large Diffeomorphic Distance Metric Mapping (LDDMM), DARTEL, diffeomorphic demons and symmetric normalization [23]–[26]. Recently, the VoxelMorph algorithm [27] used a deep neural network approach to demonstrate comparable performance on nonlinear registration according to Dice scores. The VoxelMorph architecture applies a UNet network [28] on a pair of images – the reference and the image to be warped – to predict a map of voxel-wise deformation coefficients. The voxel-wise deformation map is sent to a spatial transformer network (STN) to complete the image warping task. Gradients for training are taken over the cascade of UNet and STN on the warped image and further regularization is introduced to ensure smoothness in the deformation map. VoxelMorph establishes deep neural networks as a compelling alternative for nonlinear image registration because with GPU-enabled inference, computation time can be reduced by almost 4 orders of magnitude as compared to energy function-based optimization approaches.

As neuroscientific study expands towards multi-center studies with increasingly larger global datasets with tens of thousands of subjects [29], the sufficiency of these templates should be re-examined since many templates were developed from a small population of hundreds of subjects with limited ethnic and environmental diversity. Full automation of brain computational processing pipelines remain elusive and investigators continue to manually edit algorithm parameters on outlier datasets [30]. These deficiencies are because template-based features are based on heuristics from human experts and bears similarities with the threshold selection problem of machine vision algorithms prior to the advent of deep neural networks. The unsupervised feature learning properties of GAN could offer a valuable solution, just as deep neural network classification has solved the feature representation problem of traditional image classification tasks.

## II. RELATED WORK

Early GANs were difficult to train towards successful convergence due to poor gradient behavior [31], [32]; discontinuous or vanishing gradients at early stages of training impeded convergence to an accurate model, whereas exploding gradients at later stages of training restricted learning to a few dominant modes only. Both accuracy and diversity of representation remain important measures of GAN model performance. Both are easy to assess qualitatively, by visualizing the images synthesized from the GAN using a random set of latent vector inputs. Training with adversarial loss alone appears insufficient to guarantee reliable convergence to an accurate and diverse model. Successful image GANs also employ a convolutional architecture with batch normalization [33] and additional regularization on image content, typically an L1 or L2 loss [34] just like for autoencoder training. The regularization term appears to enhance the learning of low-pass image features while the adversarial loss is believed to encourage learning of the residual high-frequency features.

It remains challenging to assign numerical scores to the accuracy and diversity of GAN to allow fair comparison between models. The difficulty is in finding a conceptual approach to quantify performance over unknown classes and could be illustrated with the following example. For a GAN trained to synthesize images of insects, the diversity achieved cannot be quantified neatly because there are many unknown species of insects. Neither could the accuracy be computed exactly because all possibilities of natural color, size and shape variations are also unknown. Attempts were made to quantify the similarity of GAN images to a training set by passing the synthesized images through a pre-trained deep neural network and evaluating the statistics of intermediate feature maps in the network. The Inception Score [35] was computed from feature maps of the Inception Network, pre-trained on the ImageNet dataset, whereas the Mode Score [36] utilized feature maps from a specially trained discriminator network. The Frechet Inception Distance [37] compares the feature maps of the Inception v3 network from real and synthesized images, by evaluating the Frechet distance based on the assumption that the feature map obeyed multidimensional Gaussian distributions. Since deep neural network models are excellent at extracting salient image features, comparing the statistics of their feature maps should be indicative of the similarity between images. In a comparative study of GAN metrics [38], the Maximum Mean Discrepancy (MMD) [39] on feature maps of a ResNet-34 network pre-trained on ImageNet and the Frechet Inception Distance were empirically found to perform well; the MMD was recommended due to its computational simplicity. However, the Inception and Mode scores could not reliably distinguish poorly trained GANs i.e. networks that simply memorized training samples. Aside from scores derived from feature maps of deep networks, the diversity of GANs have also been empirically measured with the Multi-Scale Structural Similarity Index Measure (MS-SSIM) [40], which is an established perceptual structural similarity score borrowed from the image compression community [41].

Exploratory studies using GAN to model brain images have mostly been motivated by applications to brain tumor imaging. Initially, the GAN paradigm appeared as an interesting alternative to deep convolutional U-Net architectures for tumor segmentation [28]. Conditional GAN cast the tumor segmentation task as an image style transfer task [42]. GAN models also emerged as a competitive alternative to U-Net architectures for transforming brain images acquired from one instrument modality to images acquired by a different instrument but from the same brain [43], [44]. Learning cross-modal image transformation potentially enriches image feature representation and enhances the accuracy of tumor segmentation because of the additional information learned from cross-modal image correlations. Conditional GAN learned a mapping to synthesize MRI FLAIR contrast images from T1 contrast images prior to segmentation with a multimodal fusion algorithm [45]. Transformation of T2-contrast MRI to synthetic X-ray Computed

Tomography (CT) images was also achieved with cycle GAN [46]. The authors were motivated to use synthetic CT images to guide radiation treatment planning. The GAN modeled the mapping of radiation dose-dependent intensity in CT images to the qualitative contrast of T2-MRI. Recent innovations using the cycle GAN paradigm [46]–[48] removed the requirement for the different brain imaging modalities in the training dataset to be spatially registered. Using the GAN framework, tumor tissue may alternatively be conceptualized as anomalies appearing within healthy tissue. Preliminary efforts attempted to apply GAN to learn 2D brain image models from healthy adults [49], [50] as a complement to parallel efforts that learn features of tumor tissue. The standard deep convolutional GAN model [33] was applied to learn and synthesize 2D image slices [49] of healthy brains and also configured to generate new brain image stacks with arbitrarily placed tumor tissues [50]–[52] through both conditional and unconditional synthesis approaches. These studies established the feasibility of GAN for unsupervised learning of brain image models and also highlighted the significant opportunity for methodological improvements. However, synthesizing realistic high-resolution 2D MRI images beyond  $128 \times 128$  pixels remained a challenge as medical professionals could distinguish almost 1 out of every 2 synthetic images from real ones through visual assessments [52]. Interestingly, deformable registration has also been proposed as a data augmentation technique for MRI brain images [53]. Although proposed as a strategy to improve segmentation accuracy, this technique can also be viewed as a generative model of the brain if the deformation statistics were made to match the distribution of real brains in the population.

It is challenging to train good GAN models of high-dimensional brain image datasets. State-of-art solutions employ variations of divide-and-conquer strategies, typically in the form of progressive training of the underlying deep networks or dividing the feature learning task among multiple networks [54], [55]. Among the early attempts, a conditional GAN was trained to attain four-fold super-resolution on MRI anatomical images in patch-wise fashion [56]. GAN models were also trained using progressive growth strategies like Laplacian GAN [57] and progressive GAN [52] to synthesize individual image slices with resolution up to  $256 \times 256$  pixels but these models lacked 3D cohesiveness. tSNE was used to map the space of latent vectors to identify patterns between spatially adjacent slices with limited success [52]. Recent work attempting to remedy GAN for full 3D modeling had focused their innovations on shaping the latent vector input to the generator. The alpha-GAN [58] was successfully applied to model a full 3D MRI image stack, but only up to  $64 \times 64 \times 64$  resolution [59]. To avoid mode collapse due to the additional image dimension that must be learned, a VAE-styled encoder was trained in tandem with the GAN generator and the latent codes trained adversarially against the GAN generator latent code. The latter step shapes the

feature space of the latent code into the GAN generator network and may be interpreted as a type of regularization. In another approach, latent vectors from variational autoencoder models of neighboring brain image slices were modeled with a joint Gaussian distribution. 3D coherence was maintained by sampling slice-wise latent vectors from this joint distribution during synthesis [60] but synthesized image stacks were blurry. More recently, a hierarchy of two GANs was trained where one generator learned to synthesize a full brain volume with  $64 \times 64 \times 64$  resolution while the second generator could synthesize a super-resolution  $256 \times 256$  slice from any patch within the first brain image stack [61].

In this work, we present a novel method to synthesize high-resolution 3D image stacks of T1-contrast MRI of the anatomical brain using two deep generative adversarial neural networks. We applied a divide-and-conquer strategy to generate realistic images with a resolution of  $256 \times 256 \times 256$  voxels, which is comparable to 1.5T and 3T MRI image acquisitions for clinical use. We train a first GAN (called the shape network) with an unconditional generator using the Wasserstein GAN with Gradient Penalty (WGAN-GP) method [62] to learn the 3D spatial deformation of the brain against the MNI-152 template. The discriminator was a convolutional network with instance normalization while the generator was augmented with additional skip connections. The generator learnt the deformation statistics from a normal healthy population of the Human Connectome Project dataset and synthesizes novel voxel-wise deformation fields in three translational dimensions. The deformation map is applied to warp the template brain image stack. We trained a second GAN (called the texture network) to transform individual slices of the warped template brain to add realistic contrast and cortical fold patterns. We formulated this transformation as a style transfer task and used the pix2pix GAN method [63]. The discriminator was a convolutional network that learned a patch-wise Markovian representation of local regions within a 2D slice. The conditional generator network utilized the U-Net architecture with skip connections. Our method is competitive with other state-of-the-art 3D brain MRI image synthesis methods and overcomes the difficulties of training a GAN to learn a realistic high-dimensional model from a training dataset with limited number of images. Our new approach demonstrates the feasibility of factorizing the feature modeling task to different deep networks according to meaningful types of information namely global shape and local correlations in voxel contrast. Unlike previously reported techniques in literature, our method does not perform a coarse-to-fine resolution upgrade nor do we impose any explicit regularization or structure in the latent vector space.

The rest of this paper is organized as follows: section III describes our MRI dataset, shape and texture network architectures, training, synthesis and validation steps. In section IV, the qualitative and quantitative results are presented. The findings are discussed in section V before concluding remarks in section VI.

## A. CONTRIBUTIONS

Our main contribution is a novel methodology to synthesize high dimensional ( $256 \times 256 \times 256$ ) 3D MRI images of the brain by applying a sequence of two generative deep neural network models to transform a template brain image. These deep neural networks learn generative models of the 3D deformation field and textural features from the training dataset using the GAN method. This method overcomes the difficulties of training high-dimensional image generative models using GAN and is competitive with state-of-the-art methods for learning generative models of 3D MRI brain images.

## III. METHODOLOGY

### A. NEURAL NETWORK ARCHITECTURE

For shape network, we use a deep generative neural network architecture as depicted in Fig. 1, which is modified from CNN GAN design applied to super-resolution [64]. For the discriminator, Instance Normalization is chosen [65] instead of Batch Normalization [66] to learn instance-specific mean and covariance shift and to circumvent hardware limitations on our training batch size. The final layer is substituted with a  $1 \times 1$  convolutional block, followed by leaky ReLU and two dense layers (with leaky ReLU activation in between). The discriminator outputs a scalar decision, where a large positive value indicates high confidence that the sample image is real whereas a large negative value indicates high confidence that the sample image is synthetic.

For the generator, we have total of 6 deconvolutional layers (with stride 2) for up-sampling, and 5 deconvolutional layers (unit stride) augmented with residual connections to enhance the generator's ability to learn higher levels of abstraction. Instance Normalization [65] and the Exponential Linear Unit (ELU) [67] activation function are chosen as the normalization and nonlinear activation layers respectively. The latter accelerates learning by encouraging natural gradients [67]. A scaled hyperbolic tangent nonlinearity is applied at the final layer with a trainable scalar,  $\alpha$  to saturate the output pixels between  $[-10, 10]$  according to equation (1). The generator receives a normally distributed latent vector of size 100 with zero mean and unit variance as input and generates a deformation field of size  $256 \times 256 \times 256 \times 3$  (3D deformation field for X, Y, Z directions) as output.

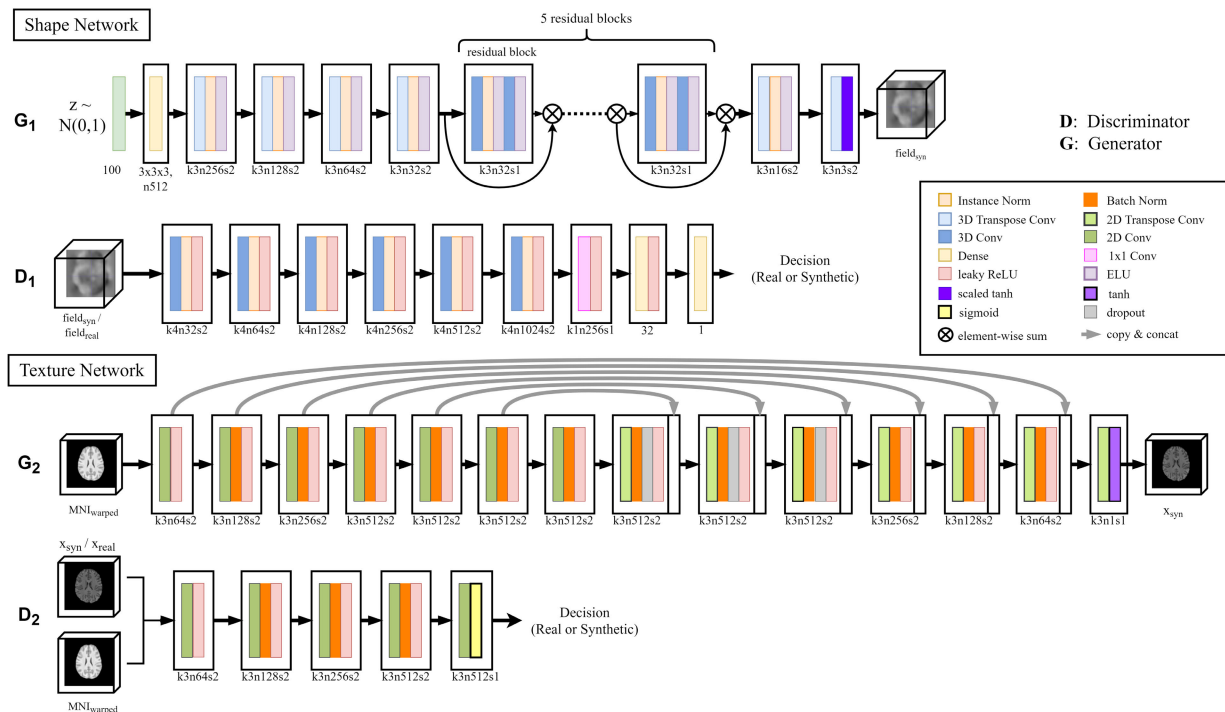
$$S(x) = \alpha \left( \frac{1}{1 - e^{-x}} - \frac{1}{2} \right) \quad (1)$$

For the texture network, we apply the pix2pix GAN [63] which consists of a U-NET [28] generator and a patch-GAN network [68] at the discriminator. The generator receives axial slices from the MNI template image stack after it has been deformed by a 3D field sampled from the shape network. It synthesizes a texturized version of the axial slice.

### B. TRAINING DATASET AND PREPROCESSING

The generative deep neural network models were trained with MRI image stacks obtained from the Human Connectome





**FIGURE 1.** Generative deep neural networks can learn salient image features from high-dimensional 3D image stacks and reliably converge to good stable solutions using the architectures depicted above. We used a discriminator-generator CNN pair to learn the models with kernel size ( $k$ ), number of channels ( $n$ ) and stride ( $s$ ) of the convolutional and dense layers indicated. Novel images are synthesized from the generator network alone. (Top) The shape network learns a 3D deformation field from a template image. (Bottom) The texture network learns a realistic mapping of voxels’ contrast and structural detail.

Project (HCP) Young Adults Brain Dataset [69]. A total of 1112 brains were skull-stripped and registered to a brain atlas (MNI ICBM 152 nonlinear 6<sup>th</sup> Generation Symmetric Average Brain Stereotaxic Registration Model) [19] using affine transformation and diffeomorphic registration, with the Symmetric Image Normalization (SyN) algorithm [26] and mutual information cost function. We used the Advanced Normalization Tools (version 2.2.0) software [22] to perform the image stack registration to the MNI template brain.

### C. TRAINING DEEP GENERATIVE NEURAL NETWORKS

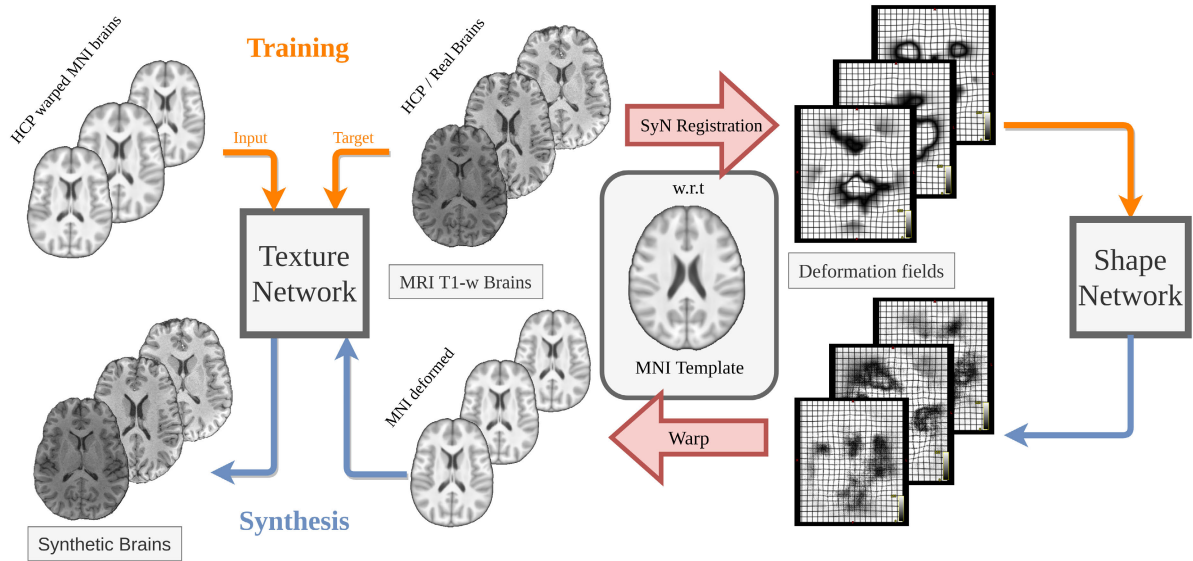
For the shape network, discriminator and generator were trained using the Wasserstein GAN with Gradient Penalty (WGAN-GP) algorithm and RMSProp [70] with a learning rate of 0.0002, momentum of 0.5, ELU’s  $\alpha$  of 1, and leaky ReLU’s  $\alpha$  of 0.2 on 200 epochs with a batch size of 2. The cost functions used for the discriminator and generator are equations (2) and (3) respectively where  $D(x)$  is the decision output of the discriminator given image  $x$  and  $G(z)$  is the image synthesized by the generator given the random latent vector  $z$ . We used default values from the paper for the number of discriminator-per-generator updates,  $k$  and regularization coefficient for the gradient penalty,  $\lambda_1$ . All weights were initialized to uniformly distributed values according to the Glorot scheme [71] and  $\alpha$  in the custom hyperbolic tangent function (1) in the generator was initialized to 20. Training

took 7-days on an NVIDIA K80 GPU.

$$L_{D1} = -\mathbb{E}_x[D(x)] + \mathbb{E}_z[D(G(z))] + \lambda_1 \mathbb{E}_{\tilde{x}}(\|\nabla D(\tilde{x})\|_2 - 1)^2 \quad (2)$$

$$L_{G1} = -\mathbb{E}_z[D(G(z))] \quad (3)$$

For the texture network, we applied the discriminator and generator cost functions from equation (4) and (5) respectively with learning rate of 0.0002, momentum  $\beta_2$  of 0.5, and regularization coefficient for the generator,  $\lambda_2$  of 100 on 100 epochs with batch size of 3. All weights were randomly initialized following a normal distribution within the range of  $[0, 0.02]$ . We inverted the deformation field of 1112 HCP brain image stacks which were obtained by SyN diffeomorphic registration to the MNI template. The deformations are used to warp the MNI image to obtain 1112 deformed MNI image stacks. All image stacks were zero-padded from an initial axial resolution of  $193 \times 229$  voxels to  $229 \times 229$  voxels. Voxel values were clipped to a maximum intensity of 900 and subsequently normalized to  $[-1, 1]$ . The image stack was then resampled using nearest-neighbor interpolation to  $256 \times 256$  voxels in axial slices. We trained the pix2pix GAN network using axial slices from the deformed MNI image stack as input and the corresponding axial slices from the true HCP image stack as the target. The network was trained on 80 axial slices from each subject. Training took 3 days on an NVIDIA



**FIGURE 2.** Graphical illustration of the workflow to train the shape and texture networks (orange arrows) and to synthesize new 3D MRI brain image stacks.

K80 GPU.

$$L_{D2} = \mathbb{E}_x \log D(x) + \mathbb{E}_y [\log 1 - D(G(y))] \quad (4)$$

$$L_{G2} = \mathbb{E}_y [D(G(y))] + \lambda_2 \|G(y) - x\| \quad (5)$$

#### D. SYNTHESIZING NOVEL BRAIN IMAGE STACKS

We sampled a 100-dimension latent vector from a zero-mean, unit-variance normal distribution as a seed to the generator of the shape network to synthesize a 3D brain deformation field. The deformation is applied to the MNI template using the Advanced Normalization Tool (version 2.2.0) software as illustrated in Fig. 2. We zero-pad to make the axial slices square and resample to  $256 \times 256 \times 256$  voxels using nearest-neighbor interpolation as previously described. All axial slices in the image stack are individually transformed by the texture network generator. Forward pass time complexity from Shape Network to Texture Network is reported to be around 0.2 seconds per instance.

#### E. ANALYZING SYNTHETIC IMAGE STACK QUALITY

To quantify the similarity of our synthetic image stacks with the HCP dataset, we use maximum mean discrepancy (MMD) [72] which is expressed as

$$\text{MMD} = \sqrt{\frac{1}{B^2} \sum g \cdot g^T + r \cdot r^T - 2r \cdot g^T}, \quad (6)$$

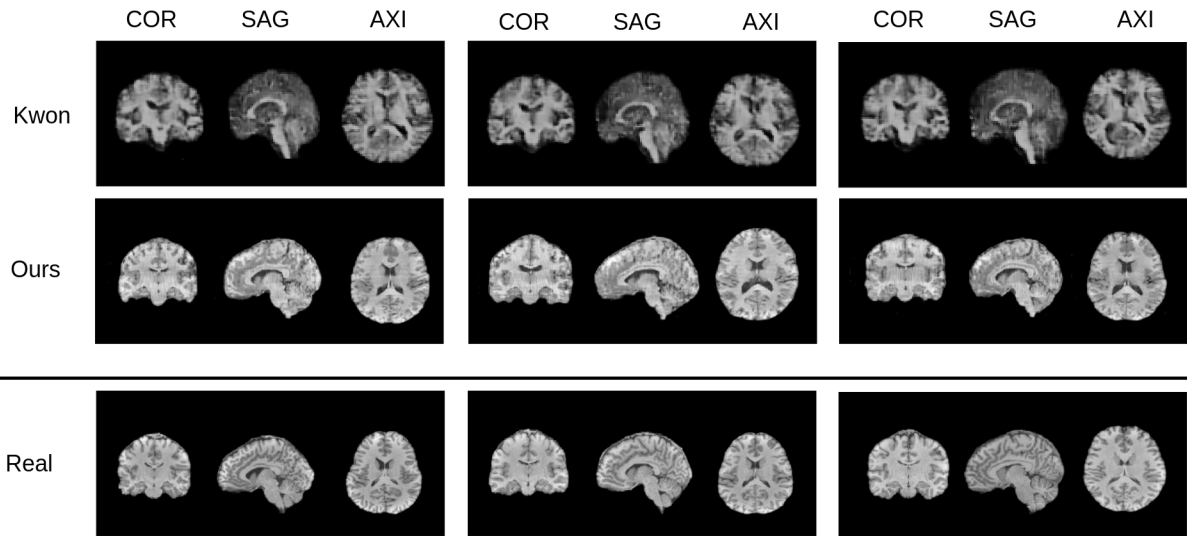
where  $g$  and  $r$  denote the flattened batch from generated samples,  $s$  and real samples,  $p$  respectively. We computed MMD over 125 tests with batch size of 8 to cover 1000 sample-pairs from the real HCP dataset and GAN-synthesized dataset. To validate our GAN model against image datasets that are not part of the training cohort, we compared the MMD score

of synthesized images from our model with 115 T1-contrast images of healthy brains from the ADNI dataset (ADNI1: Annual 2 Yr 3T) [73]. This MMD score is computed over 145 tests with batch size of 8 and labeled as “MMD (ADNI)” in Table 1.

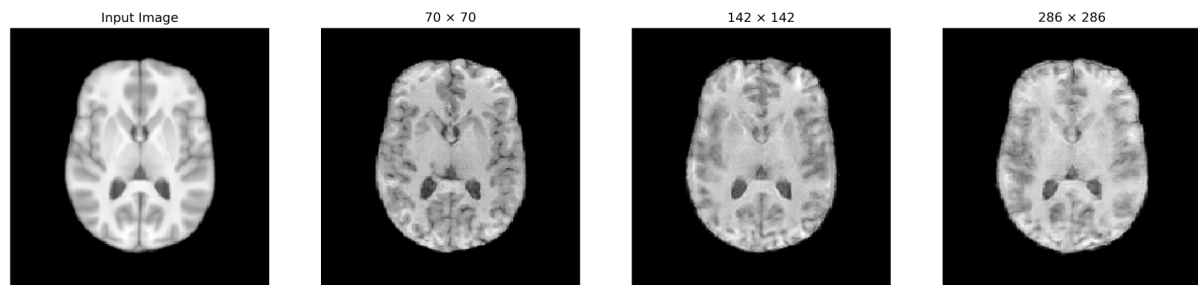
To compare the diversity of synthetic image stacks against the diversity present in the real training cohort, we computed the Multi-Scale Structural Similarity Index (MS-SSIM) [41]. The MS-SSIM is computed by comparing random, pair-wise image stack samples from the HCP or GAN-synthesized datasets. Our final MS-SSIM score was averaged over 5 tests with 1000 pair-wise comparisons per test. Both MMD and MS-SSIM scores are benchmarked with values obtained from other generative models as reported by [60, Table. 1].

To qualitatively evaluate the shape similarity of synthetic brain image stacks generated by our shape network with real shapes from the HCP dataset, we compare the probability distribution of deformation offsets from the MNI template along the X, Y, Z and radial dimensions. 1100 randomly selected image stacks from the HCP dataset were compared with 1100 synthesized brain image stacks in 8 regions of interest (ROIs): namely the whole brain, frontal lobe, temporal lobe, parietal lobe, occipital lobe, cerebellum, caudate and thalamus. The ROIs are based on MNI structural atlas [74].

The effect of patch size,  $P$  in the texture network discriminator on image quality is also evaluated at three different sizes of  $70 \times 70$ ,  $142 \times 142$  and  $286 \times 286$ . For each configuration, the network was trained from scratch while keeping the hyperparameters constant. Visual results are illustrated in Fig. 4 which show the generator’s output conditioned on the same input brain. Throughout the paper, all experiments



**FIGURE 3.** Selected coronal, sagittal and axial slices show the perceptual quality of our generative model in comparison with real images from the training dataset and an alternative neural network generative model [59].



**FIGURE 4.** Effect of varying patchGAN patch sizes on the visual quality of synthetic brains. Input image shown is deformed MNI brains after shape networks.

use a texture network with patch size of  $70 \times 70$ , unless specified.

#### IV. RESULTS

Table 1 shows a quantitative comparison of our generative model against the models of Kwon *et al.* [59] and Volokitin *et al.* [60]. Texture network with patch size of  $70 \times 70$  achieves the lowest MMD score and a competitive MS-SSIM score whereas patch sizes of  $142 \times 142$  and  $286 \times 286$  both show slight deterioration in image quality and diversity based on their respective MMD and MS-SSIM scores.

In Fig. 3, we illustrate representative axial, coronal and sagittal slices from the HCP dataset and images synthesized from Kwon *et al.* [59] and from our method. We are not able to replicate Volokitin *et al.*'s model. Our model synthesizes brain images with higher resolution at  $256 \times 256 \times 256$  voxels compared to Kwon's method at  $64 \times 64 \times 64$  voxels. Compared to state-of-the-art models, our generative model has improved perceptual and structural details such as diverse sulcal and gyral patterns. In Fig. 4, a representative axial slice shows that a texture network with patch size of  $70 \times 70$  voxels generates images with the best visual quality. We also

**TABLE 1.** MMD and MS-SSIM scores are comparative measures of accuracy and diversity of generative brain models. Patch size of the discriminator in the texture network is denoted in brackets.

| Source Images                      | MMD (ADNI) | MMD (HCP)   | MS-SSIM       |
|------------------------------------|------------|-------------|---------------|
| 3D $\alpha$ -WGAN [59]             | –          | 912627      | 0.7106        |
| 2D slice VAE [60]                  | –          | 323233      | 0.8768        |
| Shape+Texture GAN ( $70^2$ , Ours) | 1139       | <b>1009</b> | <b>0.8567</b> |
| Shape+Texture GAN ( $142^2$ )      | –          | 1042        | 0.8863        |
| Shape+Texture GAN ( $286^2$ )      | –          | 1080        | 0.8647        |
| Real HCP data                      | –          | –           | 0.7888        |
| ADNI                               | –          | –           | 0.8929        |

show visual examples of the outputs after key processing steps in Appendix VI according to the workflow depicted in Fig. 5.

We have examined the statistical distribution of deformation offsets learnt by our shape network against the distribution of deformation offsets from the HCP dataset. Fig. 6 and 7 in Appendix VI show the distributions of deformation offsets

from real and synthetic brain image stacks in the three orthogonal translational axes and in the radial dimension for 8 ROIs in the brain. The distribution derived from synthesized brain images closely approximates the distribution derived from real HCP images, which is a good indicator that the shape network has learnt gross shape reasonably well.

## V. DISCUSSION

Generative image models learn the underlying statistical distributions of complex anatomical patterns and can synthesize new examples similar in nature to the learned cohort. Generative models that are capable of successful and realistic image synthesis are also models that have learned effective representations of the salient anatomical features of the brain. In this way, the ability to synthesize realistic images is also a key step to enhance computational image understanding. The challenge for generative modeling is for the learning algorithms to converge reliably to neural network models that achieve both high accuracy and diversity in the representation of the underlying training dataset. The difficulty is not proportionally compounded when image size grows because the generative model must learn not just the additional information content of the extra voxels but also the correlations with other voxels. Training generative models shares similarities with solving an underdetermined system of equations. To encourage convergence towards good models, additional constraints must be introduced via regularization. We believe it is the lack of good regularization paradigms that is the key limiting factor precluding successful and reliable training of GAN models for high-resolution 3D image volumes thus far. Techniques like super-resolution GAN and progressive GAN appear as effective regularizers that enable GAN models to learn high-resolution 2D image models but appear insufficient for more complex tasks like image inpainting and generative modeling of image volumes with a 3<sup>th</sup> dimension and beyond.

Our approach is also a form of regularization but not via explicit constraint on the loss function. We train two models to learn more specific and restricted types of information namely the shape deformation and a style mapping of local intensity correlations applied to a template brain rather than a full model of 3D voxel intensity values. Learning the voxel identities alone is a much more difficult problem as the voxel representation must encode the spatial relationship between brain structures as well as all positional, textural and shape-related features of each region-of-interest (ROI). Learning both a deformation field from the MNI template and a textural mapping from the deformed template simplifies the problem to a lower-dimensional learning task; this helps high-dimensional GAN converge to good solutions using a limited dataset. Our generative models learn only the distribution of shape and textural perturbations from a template brain. The generative shape model only learns the encoding of orientation, translation, shape and size of brain ROIs whereas the spatial arrangement of ROIs is encoded in the MNI template brain. The generative texture model only

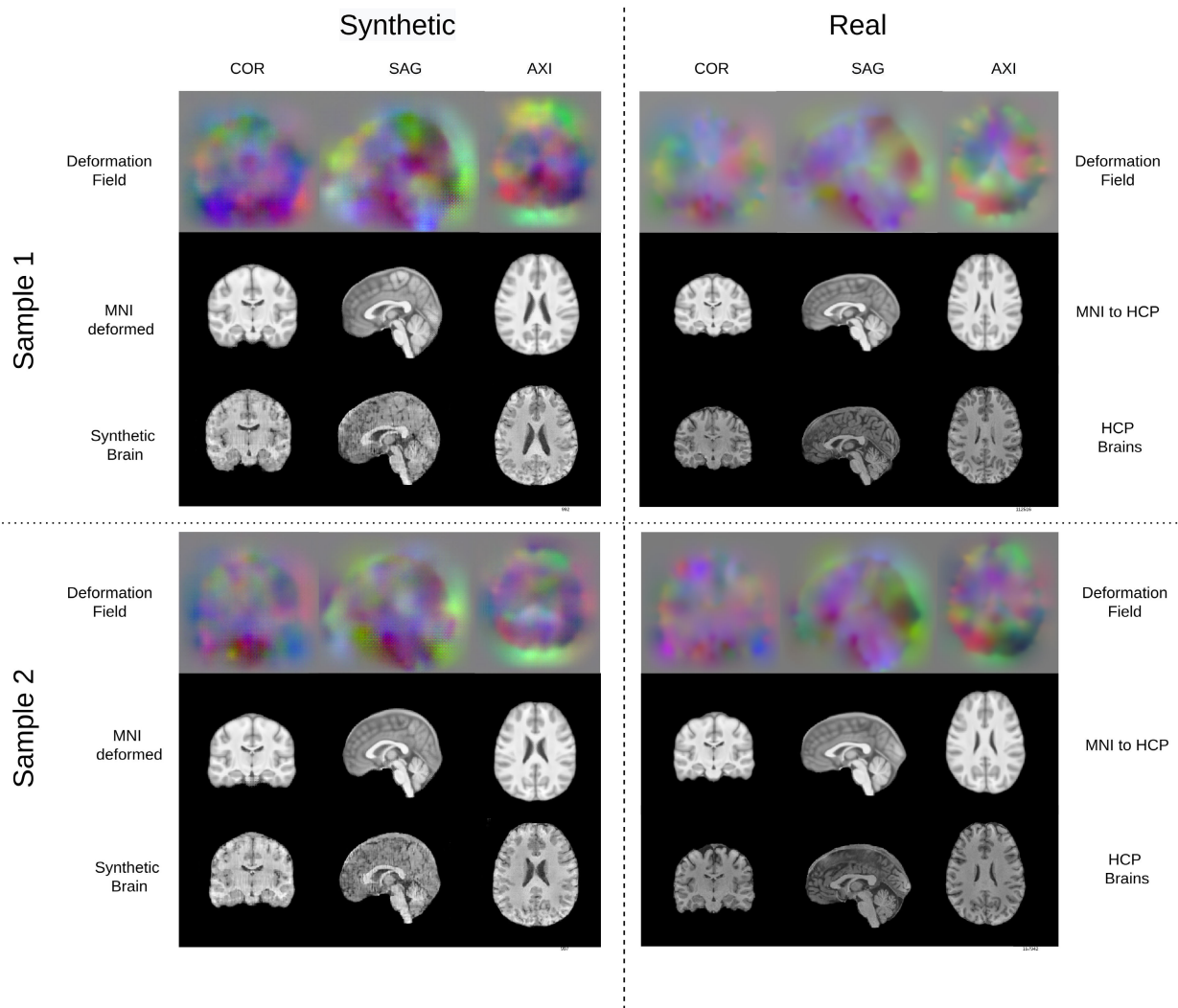
learns the mapping of ROI-dependent voxel values but the localization and clustering of ROI-related voxels are inherent in the MNI template brain. This strategy ultimately allows us to combine two generative models to synthesize a full 3D brain image stack with realistic voxel values at high image resolution.

Our model accuracy is assessed by comparing the MMD scores of the synthesized image stacks against real brain image stacks from the HCP dataset. Lower values of the MMD metric indicate a greater similarity of voxel distributions and imply that synthesized images are visually similar to the training dataset. The model diversity can be estimated by comparing the MS-SSIM scores of synthesized image stacks against the HCP dataset. Lower values of MS-SSIM indicate that samples are diverse whereas high values of MS-SSIM suggest that the generative network suffers from mode collapse, a common failure of GAN networks where only a single or very few unique images are synthesized. Generated samples should have similar MS-SSIM scores to the real dataset as this shows that the synthesized image stacks have similar levels of diversity with the empirical dataset. Compared to state-of-art generative 3D MRI brain models, our shape and texture GAN model achieves the best MMD score. This comparison suggests that our texture network has effectively learned realistic voxel intensity values and realistic variations of the sulcal and gyral patterns of the cortex. Our model also has a competitive MS-SSIM score which indicates slightly less diversity than the real dataset. To validate our model against image datasets that are not part of the training cohort, we compared the MMD score of synthesized images from our model with 115 images of healthy brains from the ADNI dataset [73].

The results presented in Table 1 show minimal difference between MMD (ADNI) and MMD (HCP) scores which support our claim that our GAN networks have learned salient features which appear indistinguishable from the validation dataset from ADNI. While MMD and MS-SSIM measurements allow convenient comparison between GAN models and serve as useful quantitative feedback to gauge improvements in methodology, these metrics cannot be interpreted as absolute measures of performance. This work still lacks perceptual evaluation from radiologists. The GAN model would appear to have learnt clinically important structural features only if the synthesized images cannot be differentiated from real brain images by radiologists.

To describe our innovation, it is informative to interpret our method through the lens of three different paradigms and contrast with work reported from literature. The first paradigm uses the concept of progressively upgrading synthesized images from a lower-quality estimate to high-quality output. This paradigm is best illustrated by the work of Sun [61] and thus far, quality upgrades have been approached from the perspective of enhancing image resolution. Our work proposes a different approach where the texture network upgrades the quality of local voxel contrast and correlations which we have framed as a style transfer task. Viewed from



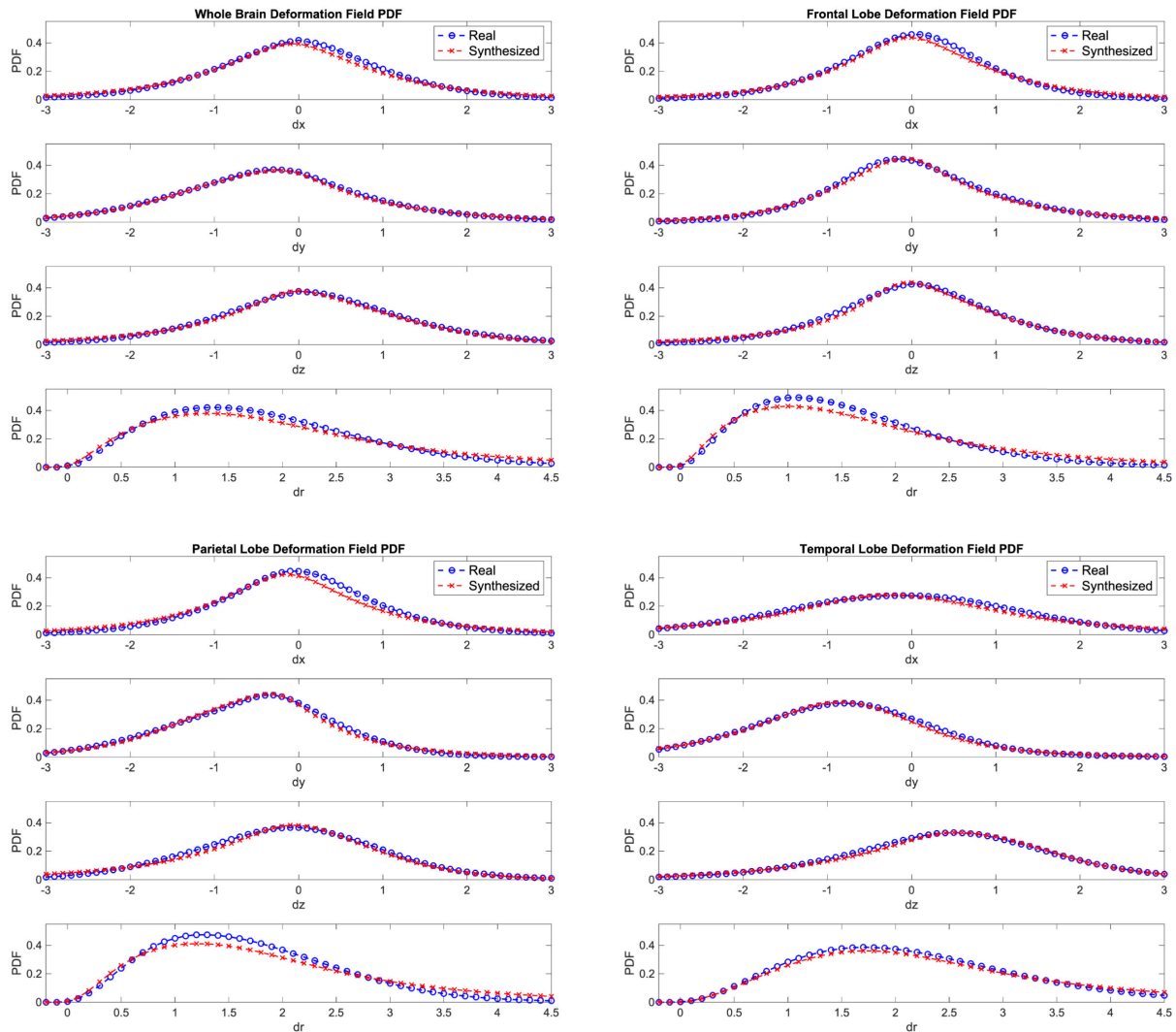


**FIGURE 5.** Samples of deformation field (in RGB display), MNI deformed brain, and synthetic brains compared to real samples at the same spatial location. Displacement in each spatial dimension is mapped to each RGB color channels for deformation field.

this perspective, the model learnt by the patch-wise discriminator of our texture network is critically important as it should learn a Markov Random Field representation of the local voxel intensity correlations. For brain image stacks, the optimal size of the local neighborhood to capture correlations is unknown. We conducted experiments with three different patch sizes ( $70 \times 70$ ,  $142 \times 142$  and  $286 \times 286$ ) and observed that spatial sharpness decreased as patch sized increased. As illustrated in Figure 4, we selected a patch size of  $70 \times 70$  as it yielded the sharpest features in synthesized images. The second paradigm is about shaping the latent vector space as a form of regularization, by imposing additional constraints on the random vector input into the unconditional GAN generators. It has been applied as a regularizer to promote convergence towards a good model [59] during GAN training and to enforce structural consistency of synthesized image slices across the 3<sup>th</sup> image dimension [60]. In contrast to the literature, our method can be interpreted as applying

additional constraints (regularization) on the latent vector of a conditional GAN generator i.e the texture network. The shape network acts as a mapping function to apply the constraints of realistic variations in neuroanatomical shape. Viewed from this perspective, the shape network could be further improved towards a deep neural network-only solution by adapting ideas from the VoxelMorph [27] forward deformation map and spatial transformer networks.<sup>1</sup> The third paradigm takes the view of applying multiple deep networks to learn and synthesize different aspects of the information contained in MRI brain images. We have not found illustrative examples applied to MRI brain image modeling but StyleGAN [7] and its variants [8] embody this idea. StyleGAN introduced innovations to the generator where several subnetworks with different architectures collaborate to factorize the generative model into controllable parameters. The exciting result was

<sup>1</sup>We thank the anonymous reviewer who recommended this paper to us.

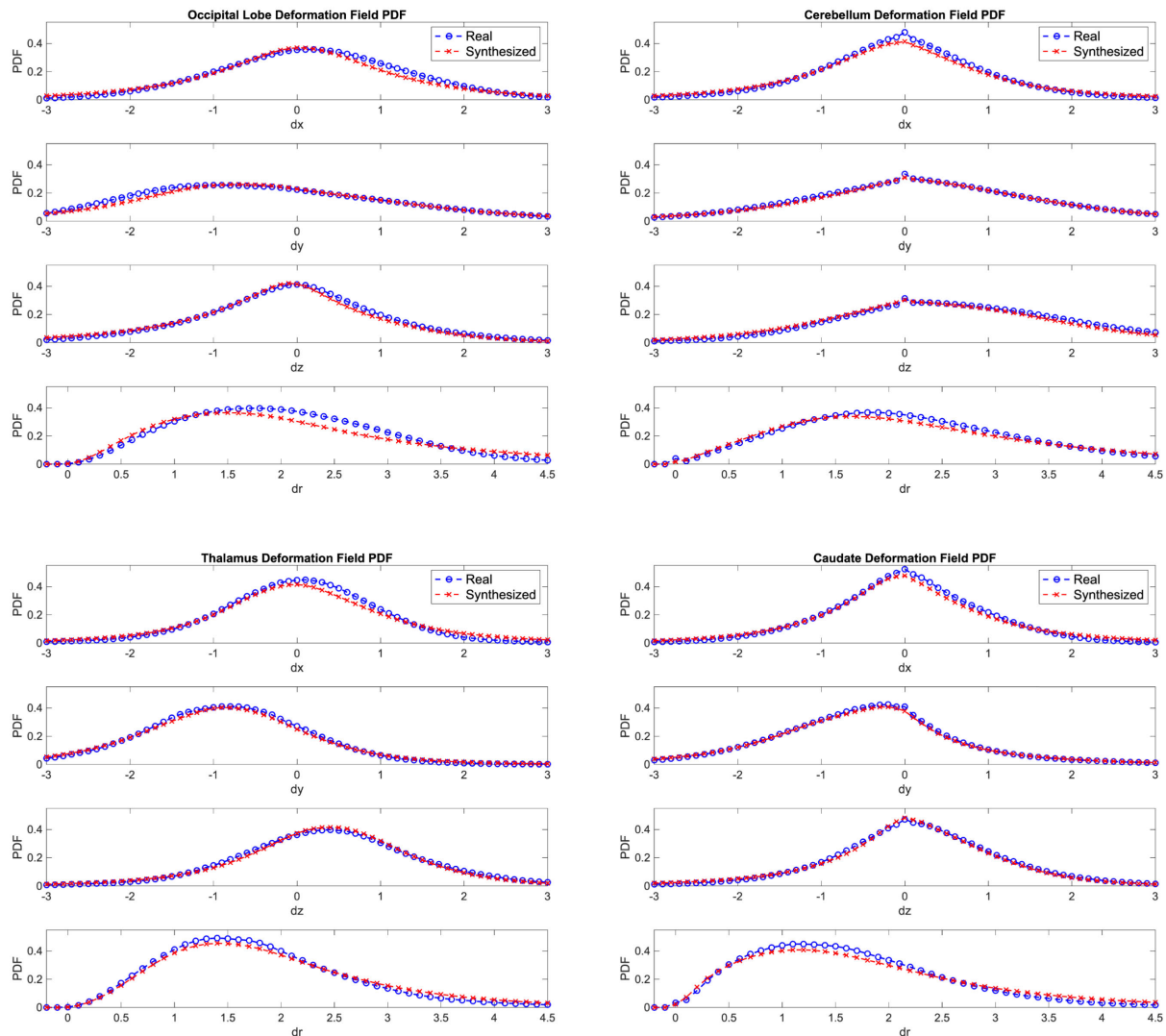


**FIGURE 6.** Deformation Field PDF of 1100 of real and synthesized brains in whole brain, frontal lobe, temporal lobe and parietal lobe (clockwise direction) where PDF of  $dx$ ,  $dy$ ,  $dz$  and  $dr$  channels are shown in each subplot.

that the unsupervised factorization appeared to learn explainable parameters such as orientation and parts of the object in the images. Here we implement a supervised factorization instead, by explicitly assigning the type of information content to be learned by the shape and texture networks respectively. Future adaptations of our method may evolve towards more unsupervised factorization of information representation between multiple different GAN models.

Our approach is different but not mutually exclusive to the previously reported methods of learning generative models of high-dimensional images. Therefore, there is potential to integrate the best ideas from literature with our technique to further enhance the representation power of our GAN. For instance, applying latent vector shaping strategies to the conditional texture network could likely improve the 3D spatial coherence of the sulcal and gyral patterns produced. Another interesting possibility would be to enhance

our shape network to learn finer deformation features of the sulcal and gyral folds perhaps by adapting the deformation and spatial transformer deep networks or by developing another patch-wise GAN to learn only fine deformation features. Energy function-based registration methodologies are not able to capture the fine spatial variations at the level of sulci and gyri so it remains an interesting challenge to see if deep neural network approaches can outperform in this task. It is also possible to apply our GAN shape and texture approach to model separate region-of-interest (ROI) with distinct networks rather than the whole-brain modeling approach presented in this paper. The ROI level models could be combined by adapting deep network techniques from data augmentation approaches for image segmentation [53]. This potential approach is akin to multi-scale modeling and anticipated to further improve regularization because there would be significantly less variation in shape and local voxel



**FIGURE 7.** Deformation Field PDF of 1100 real and synthesized brains in occipital lobe, cerebellum, caudate and thalamus (clockwise direction) where PDF of  $dx$ ,  $dy$ ,  $dz$  and  $dr$  channels are shown in each subplot.

intensity correlations at the ROI level since ROI are typically delineated by neuronal cells with similarity in MRI contrast. Our work is most similar in conceptual approach with Zhao [53] who also developed a deep neural network approach to model spatial and appearance transformations to generate novel MRI brain image samples. This work is based on the U-Net architecture and takes a supervised learning approach whereas our approach utilizes the unsupervised GAN methodology. Interestingly, we have noted that tasks utilizing U-net architectures can also be mapped to GAN networks and this work further hints at a possible duality between supervised training of U-Net architectures to unsupervised training via GAN. It is an advantage of our method that labeled ROI segments of the brain are not required for training. However, when such labeled data is also available, we believe it would be beneficial to explore how ideas proposed by Zhao to model ROI-level detail could be adapted to

improve our whole-brain model, without the need to create separate GAN for each ROI.

In this work, we develop GAN models of healthy brains from anatomical MRI images as a first step towards a larger, future objective of applying the improved feature learning properties of generative modeling to next-generation image understanding tasks. We are motivated by three future use-cases of GAN brain image models. First, GAN models have the potential to redefine compression and storage of brain image stacks, where generative models may not only yield feature-preserving compression but also incorporate corrections to instrument non-idealities, sidestep compute-intensive tomographic or k-space reconstructions, or combined with new paradigms of pulse sequence scanning to yield substantially smaller data acquisition and faster MRI scans without compromising information content. Second, GAN models can enable a new approach to high-dimensional

image comparison not based on Fisher's statistical significance testing, which has limitations on multiple testing for high dimensional datasets. We foresee future innovations where GAN will be used as a tool to learn a high dimensional probability distribution on an image dataset. To compare two populations, for example a healthy cohort and a diseased cohort, GAN models of both cohorts could be compared with either the Fisher Information Matrix or Wasserstein distance (Earth mover Distance). Information geometry literature highlights these two distance measures as proper metric functions on the space parameterized by probability distributions that give a fair and objective measure of difference. In theory, it is possible to compute both metrics from GAN models whereas in practice, research is ongoing to improve the computational tractability. Nevertheless, the mathematics of information geometry and Frechet statistics provide a convenient theoretical framework to compare high dimensional datasets in the space of probability distributions and thereby avoid the difficulties of multiple comparisons when comparing MRI brain image stacks in voxel-space. Finally, the generator of GAN brain models may be used in neuroanatomy education especially for radiology and neurosurgery training. Utilizing existing brain processing pipelines, 3D virtual reality models can be extracted from synthetically generated MRI T1-contrast image stacks. These graphical models can help medical trainees learn the range of possible anatomical variations that may be encountered in human subjects or train surgical robots to perform automated surgeries.

## VI. CONCLUSION

Realistic image generators are a key step towards enhancing computational image understanding for automated medical diagnostics especially for complex organs like the brain. We report a novel method for learning high-resolution generative models of brain MRI image stacks using a combination of deformation field and texture transformation deep neural networks. Sharing the generative modeling between two distinct networks is a regularization strategy that enables us to train the networks on high-resolution images towards stable convergence using generative adversarial network methods. Our method can generate realistic and diversified brain image stacks that outperform state-of-the-art methods on the maximum mean discrepancy (MMD) metric with a competitive multi-scale structural similarity index (MS-SSIM) score. One limitation of this method is the cohesiveness of the sulcal and gyral folds in the axes orthogonal to axial slices, where a novel approach is possibly required to improve the fidelity of feature representation.

## ADDITIONAL SAMPLES

See Figure 5.

## DEFORMATION FIELD PROBABILITY DENSITY FUNCTION

See Figures 6 and 7.

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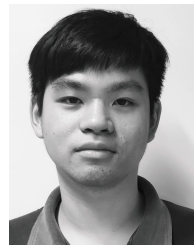
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